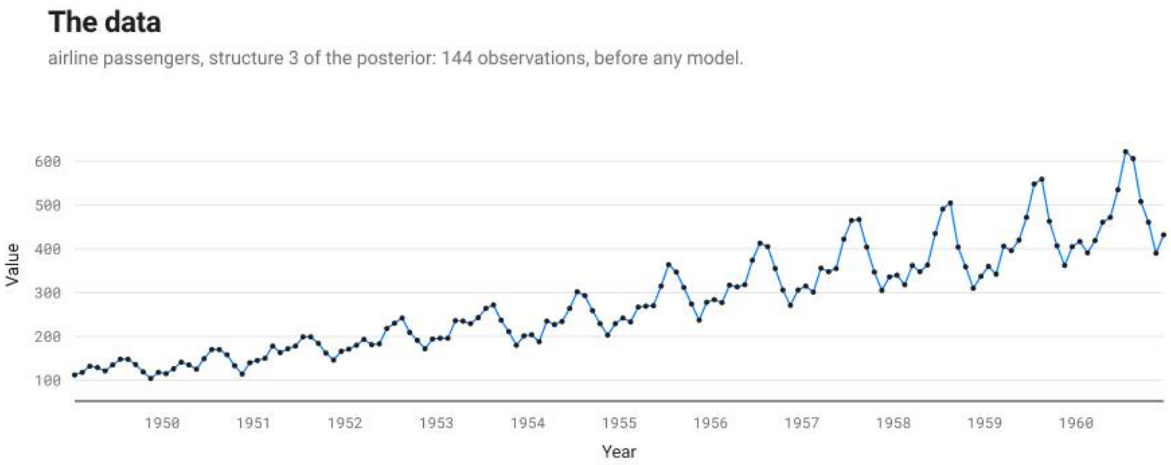
An automatic report for the dataset: airline passengers, structure 3 of the posterior

`autostat` (automatic-statistician-revisited)

08 November 2026

Produced by ***`autostat`****, a Python 3 port of the Automatic Bayesian Covariance Discovery system of Lloyd et al. (AAAI 2014).*



The data

Executive summary

With posterior weight 0.089, AutoGP kept `WN + (C + Lin + Per + GE x (C + Lin) x (C + GE + Lin)) x (C + Lin) x (C + Lin)` (Bayesian information criterion 1221.6; negative log likelihood 558.6 on 144 points). It decomposes into 37 additive components, listed by how much each reduces extrapolation error:

1. A linearly increasing function.
2. A linearly increasing function.
3. A quadratic polynomial.
4. A quadratic polynomial.
5. A linearly increasing function.
6. A quadratic polynomial.
7. A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Nov 1951.

8. A periodic function with a period of 1.0 years.
9. A rough function with marginal standard deviation increasing linearly away from Feb 1950.
10. A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Jun 1954.
11. A periodic function with a period of 1.0 years but with quadraticly varying amplitude.
12. A rough function.
13. A rough function with quadraticly varying marginal standard deviation.
14. A rough function with marginal standard deviation increasing linearly away from Nov 1951.
15. A rough function.
16. A rough function with marginal standard deviation increasing linearly away from Nov 1951.
17. A rough function with quadraticly varying marginal standard deviation.
18. A rough function with marginal standard deviation increasing linearly away from Dec 1951.
19. A rough function with marginal standard deviation increasing linearly away from Dec 1951.
20. A rough function with quadraticly varying marginal standard deviation.
21. A rough function with quadraticly varying marginal standard deviation.
22. Uncorrelated noise.
23. A rough function with cubicly varying marginal standard deviation.
24. A rough function with quadraticly varying marginal standard deviation.
25. A rough function with cubicly varying marginal standard deviation.
26. A rough function with quadraticly varying marginal standard deviation.
27. A rough function with quadraticly varying marginal standard deviation.
28. A rough function with quarticly varying marginal standard deviation.
29. A rough function with cubicly varying marginal standard deviation.
30. A rough function with quadraticly varying marginal standard deviation.
31. A rough function with cubicly varying marginal standard deviation.
32. A rough function with marginal standard deviation increasing linearly away from Jun 1954.
33. A rough function with quadraticly varying marginal standard deviation.
34. A rough function with cubicly varying marginal standard deviation.

35. A rough function with marginal standard deviation increasing linearly away from Jun 1954.

36. A cubic polynomial.

37. A constant.

#	R^2 cumulative (%)	Residual variance explained (%)	CV MAE	MAE reduction (%)
data			280	
1	32.4	32.4	247	12.0
2	63.3	45.7	219	11.1
3	76.9	37.0	200	8.7
4	80.0	13.6	195	2.5
5	86.2	30.9	191	2.1
6	86.1	-0.8	189	1.2
7	92.8	48.3	189	0.2
8	98.2	75.6	188	0.1
9	98.6	18.1	188	0.1
10	99.2	42.1	188	0.1
11	99.5	40.3	188	0.1
12	99.7	33.1	188	0.0
13	99.7	21.3	188	0.0
14	99.8	8.9	188	0.0
15	99.8	7.9	188	0.0
16	99.8	2.0	188	0.0
17	99.8	3.3	188	0.0
18	99.8	1.3	188	0.0
19	99.8	0.3	188	0.0
20	99.8	0.7	188	0.0
21	99.8	0.2	188	0.0
22	99.9	43.7	188	0.0
23	99.9	0.0	188	-0.0
24	99.9	0.1	188	-0.0
25	99.9	0.1	188	-0.0
26	99.9	0.2	188	-0.0
27	99.9	0.3	188	-0.0
28	99.9	0.2	188	-0.0
29	99.9	0.5	188	-0.0
30	99.9	1.1	188	-0.0
31	99.9	1.8	188	-0.0
32	99.9	0.5	188	-0.0
33	99.9	4.6	188	-0.0
34	99.9	1.3	188	-0.0
35	99.9	0.5	188	-0.0
36	100.0	100.0	190	-1.5
37	100.0	2.9	281	-47.4

MAE: mean absolute error over ten contiguous held-out blocks; CV: cross-validated.

The formula found

The structure search is a symbolic regression: it searches a grammar of expressions over kernels and keeps the one with the lowest Bayesian information criterion. This is the expression it kept, with every fitted parameter in place; t is in years.

$$\begin{aligned} y(t) = 280.3 &+ f_1(t) + f_2(t) + f_3(t) + f_4(t) + f_5(t) + f_6(t) + f_7(t) + f_8(t) + f_9(t) + f_{10}(t) \\ &+ f_{11}(t) + f_{12}(t) + f_{13}(t) + f_{14}(t) + f_{15}(t) + f_{16}(t) + f_{17}(t) + f_{18}(t) + f_{19}(t) \\ &+ f_{20}(t) + f_{21}(t) + f_{22}(t) + f_{23}(t) + f_{24}(t) + f_{25}(t) + f_{26}(t) + f_{27}(t) + f_{28}(t) \\ &+ f_{29}(t) + f_{30}(t) + f_{31}(t) + f_{32}(t) + f_{33}(t) + f_{34}(t) + f_{35}(t) + f_{36}(t) + f_{37}(t) \end{aligned}$$

Each component is a Gaussian process with its own covariance:

$$f_1(t) \sim \mathrm{GP}\big(0,\ \ 107.1\,(t-1950)(t'-1950)\big)$$

$$f_2(t) \sim \mathrm{GP}\big(0,\ \ 110.3\,(t-1952)(t'-1952)\big)$$

$$f_3(t) \sim \mathrm{GP}\big(0,\ \ 1.394\,(t-1950)(t'-1950)\cdot(t-1952)(t'-1952)\big)$$

$$f_4(t) \sim \mathrm{GP}\big(0,\ \ 1.025\,(t-1954)(t'-1954)\cdot(t-1950)(t'-1950)\big)$$

$$f_5(t) \sim \mathrm{GP}\big(0,\ \ 81.05\,(t-1954)(t'-1954)\big)$$

$$f_6(t) \sim \mathrm{GP}\big(0,\ \ 1.055\,(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\big)$$

$$f_7(t) \sim \mathrm{GP}\left(0,\ \ 14.13\,(t-1952)(t'-1952)\cdot\frac{\exp\Big(4.213\cos\frac{2\pi(t-t')}{0.9981}\Big)-I_0(4.213)}{e^{4.213}-I_0(4.213)}\right)$$

$$f_8(t) \sim \mathrm{GP}\left(0,\ \ 1085\,\frac{\exp\Big(4.213\cos\frac{2\pi(t-t')}{0.9981}\Big)-I_0(4.213)}{e^{4.213}-I_0(4.213)}\right)$$

$$f_9(t) \sim \mathrm{GP}\left(0,\ \ 1.114\,\exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot(t-1950)(t'-1950)\right)$$

$$f_{10}(t) \sim \mathrm{GP}\left(0,\ \ 10.38\,(t-1954)(t'-1954)\cdot\frac{\exp\Big(4.213\cos\frac{2\pi(t-t')}{0.9981}\Big)-I_0(4.213)}{e^{4.213}-I_0(4.213)}\right)$$

$$f_{11}(t) \sim \mathrm{GP}\left(0,\ \ 0.1352\,(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\cdot\frac{\exp\Big(4.213\cos\frac{2\pi(t-t')}{0.9981}\Big)-I_0(4.213)}{e^{4.213}-I_0(4.213)}\right)$$

$$f_{12}(t) \sim \mathrm{GP}\left(0,\ \ 27.01\,\exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot\exp\left(-\left(\frac{|t-t'|}{13.16}\right)^{0.5159}\right)\right)$$

$$f_{13}(t) \sim \mathrm{GP}\left(0,\ 0.01451\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot(t-1950)(t'-1950)\cdot(t-1952)(t'-1952)\right)$$

$$f_{14}(t) \sim \mathrm{GP}\left(0,\ 0.3516\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot\exp\left(-\left(\frac{|t-t'|}{13.16}\right)^{0.5159}\right)\cdot(t-1952)(t'-1952)\right)$$

$$f_{15}(t) \sim \mathrm{GP}\left(0,\ 7.358\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\right)$$

$$f_{16}(t) \sim \mathrm{GP}\left(0,\ 0.0958\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot(t-1952)(t'-1952)\right)$$

$$f_{17}(t) \sim \mathrm{GP}\left(0,\ 0.002485\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot(t-1950)(t'-1950)\cdot(t-1952)(t'-1952)\right)$$

$$f_{18}(t) \sim \mathrm{GP}\left(0,\ 0.06023\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot\exp\left(-\left(\frac{|t-t'|}{13.16}\right)^{0.5159}\right)\cdot(t-1952)(t'-1952)\right)$$

$$f_{19}(t) \sim \mathrm{GP}\left(0,\ 0.01641\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot(t-1952)(t'-1952)\right)$$

$$f_{20}(t) \sim \mathrm{GP}\Bigg(0,\ 7.842 \times 10^{-4}\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot\exp\left(-\left(\frac{|t-t'|}{13.16}\right)^{0.5159}\right)\cdot(t-1952)(t'-1952)\cdot(t-1952)(t'-1952)\Bigg)$$

$$f_{21}(t) \sim \mathrm{GP}\left(0,\ 2.136\times10^{-4}\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot(t-1952)(t'-1952)\cdot(t-1952)(t'-1952)\right)$$

$$f_{22}(t) \sim \mathrm{GP}(0,\ 30.7\,\delta_{tt'})$$

$$f_{23}(t) \sim \mathrm{GP}\Bigg(0,\ 2.044 \times 10^{-6}\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\cdot(t-1952)(t'-1952)\Bigg)$$

$$f_{24}(t) \sim \mathrm{GP}\left(0,\ 1.57\times10^{-4}\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\right)$$

$$f_{25}(t) \sim \mathrm{GP}\Bigg(0,\ 7.504 \times 10^{-6}\ \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right)\cdot\exp\left(-\left(\frac{|t-t'|}{13.16}\right)^{0.5159}\right)\cdot(t-1954)(t'-1954)\cdot(t-1952)(t'-1952)\cdot(t-1952)(t'$$

$$f_{26}(t) \sim \mathrm{GP}\Bigg(0,\ 5.764 \times 10^{-4} \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot \exp\left(-\left(\frac{|t-t'|}{13.16}\right)^{0.5159}\right) \cdot (t-1954)(t'-1954) \cdot (t-1952)(t'-1952)\Bigg)$$

$$f_{27}(t) \sim \mathrm{GP}\left(0,\ 9.167 \times 10^{-4} \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot (t-1954)(t'-1954) \cdot (t-1952)(t'-1952)\right)$$

$$f_{28}(t) \sim \mathrm{GP}\Bigg(0,\ 3.096 \times 10^{-7} \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot (t-1950)(t'-1950) \cdot (t-1954)(t'-1954) \cdot (t-1952)(t'-1952) \cdot (t-1952)(t'-195$$

$$f_{29}(t) \sim \mathrm{GP}\Bigg(0,\ 2.378 \times 10^{-5} \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot (t-1950)(t'-1950) \cdot (t-1954)(t'-1954) \cdot (t-1952)(t'-1952)\Bigg)$$

$$f_{30}(t) \sim \mathrm{GP}\left(0,\ 0.003365 \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot \exp\left(-\left(\frac{|t-t'|}{13.16}\right)^{0.5159}\right) \cdot (t-1954)(t'-1954) \cdot (t-1952)(t'-1952)\right)$$

$$f_{31}(t) \sim \mathrm{GP}\Bigg(0,\ 3.235 \times 10^{-5} \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot (t-1950)(t'-1950) \cdot (t-1952)(t'-1952) \cdot (t-1952)(t'-1952)\Bigg)$$

$$f_{32}(t) \sim \mathrm{GP}\left(0,\ 0.07041 \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot (t-1954)(t'-1954)\right)$$

$$f_{33}(t) \sim \mathrm{GP}\left(0,\ 0.01066 \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot (t-1950)(t'-1950) \cdot (t-1954)(t'-1954)\right)$$

$$f_{34}(t) \sim \mathrm{GP}\Bigg(0,\ 1.388 \times 10^{-4} \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot (t-1950)(t'-1950) \cdot (t-1954)(t'-1954) \cdot (t-1952)(t'-1952)\Bigg)$$

$$f_{35}(t) \sim \mathrm{GP}\left(0,\ 0.2585 \exp\left(-\left(\frac{|t-t'|}{0.6954}\right)^{1.031}\right) \cdot \exp\left(-\left(\frac{|t-t'|}{13.16}\right)^{0.5159}\right) \cdot (t-1954)(t'-1954)\right)$$

$$f_{36}(t) \sim \mathrm{GP}\left(0,\ 0.01334\,(t-1950)(t'-1950) \cdot (t-1954)(t'-1954) \cdot (t-1952)(t'-1952)\right)$$

$$f_{37}(t) \sim \mathrm{GP}(0,\ \ 8469)$$

Polynomial and periodic components have a closed form, read from their posterior mean:

$$f_1(t) = -8.199 + 6.779\,(t - 1949)$$

$$f_2(t) = -25.29 + 8.9\,(t - 1949)$$

$$f_3(t) = 2.471 - 2.912\,(t - 1949) + 0.7189\,(t - 1949)^2$$

$$f_4(t) = 2.368 - 2.392\,(t - 1949) + 0.3593\,(t - 1949)^2$$

$$f_5(t) = -47.91 + 8.794\,(t - 1949)$$

$$f_6(t) = 2.834 - 1.517\,(t - 1949) + 0.183\,(t - 1949)^2$$

$$\begin{aligned} f_8(t) \approx\ & 28.14\cos\left(\frac{2\pi\cdot(t-1949)}{0.9981} + 3.04\right) + 15.96\cos\left(\frac{2\pi\cdot 2\cdot(t-1949)}{0.9981} - 1.443\right) \\ & + 6.531\cos\left(\frac{2\pi\cdot 3\cdot(t-1949)}{0.9981} + 1.143\right) + 6.615\cos\left(\frac{2\pi\cdot 4\cdot(t-1949)}{0.9981} + 0.06367\right) \\ & + 3.463\cos\left(\frac{2\pi\cdot 5\cdot(t-1949)}{0.9981} - 0.4439\right) + 0.2155\cos\left(\frac{2\pi\cdot 6\cdot(t-1949)}{0.9981} + 1.217\right) \\ & + 0.6769\cos\left(\frac{2\pi\cdot 7\cdot(t-1949)}{0.9981} + 2.257\right) + 0.1764\cos\left(\frac{2\pi\cdot 8\cdot(t-1949)}{0.9981} + 1.745\right) \end{aligned}$$

$$f_{36}(t) = 0.4151 - 0.5655\,(t - 1949) + 0.2106\,(t - 1949)^2 - 0.02217\,(t - 1949)^3$$

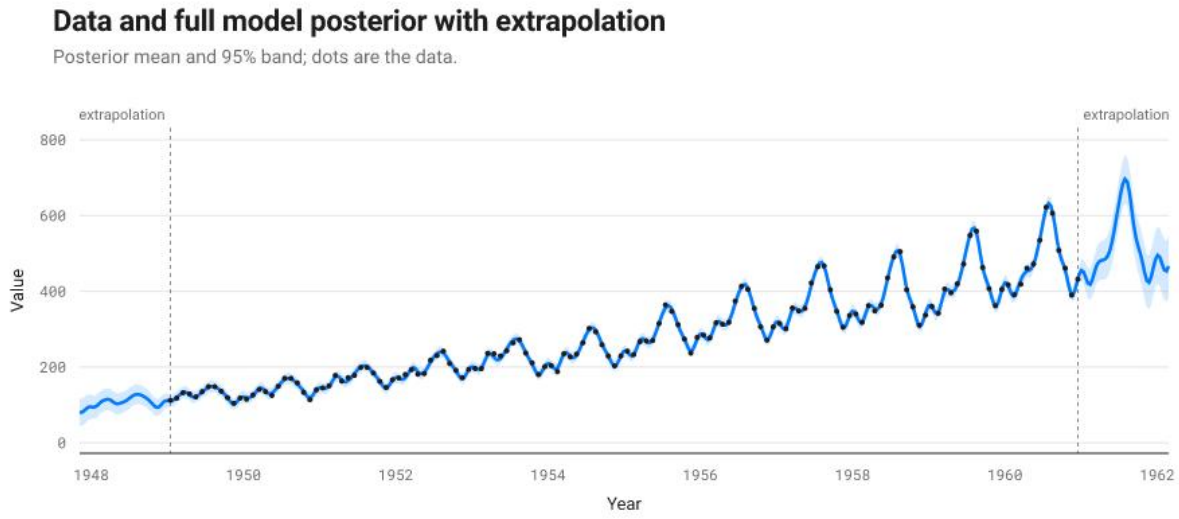
$$f_{37}(t) = -90.33$$

A periodic kernel is exactly a sum of cosines, so a periodic component is a Fourier series in its period; it is written up to the harmonic after which the rest weighs less than 0.1

The other components are smooth, quasi-periodic or noise: a draw from such a process has no finite closed form; its covariance above is the formula.

Closed form of each component:

#	component	closed form
1	A linearly increasing function	exact polynomial of degree 1
2	A linearly increasing function	exact polynomial of degree 1
3	A quadratic polynomial	exact polynomial of degree 2
4	A quadratic polynomial	exact polynomial of degree 2
5	A linearly increasing function	exact polynomial of degree 1
6	A quadratic polynomial	exact polynomial of degree 2
7	A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Nov 1951	none: the covariance is the formula
8	A periodic function with a period of 1.0 years	Fourier series, 8 harmonics, largest gap 0.028 % of the component
9	A rough function with marginal standard deviation increasing linearly away from Feb 1950	none: the covariance is the formula
10	A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Jun 1954	none: the covariance is the formula
11	A periodic function with a period of 1.0 years but with quadraticly varying amplitude	none: the covariance is the formula
12	A rough function	none: the covariance is the formula
13	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
14	A rough function with marginal standard deviation increasing linearly away from Nov 1951	none: the covariance is the formula
15	A rough function	none: the covariance is the formula
16	A rough function with marginal standard deviation increasing linearly away from Nov 1951	none: the covariance is the formula
17	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
18	A rough function with marginal standard deviation increasing linearly away from Dec 1951	none: the covariance is the formula
19	A rough function with marginal standard deviation increasing linearly away from Dec 1951	none: the covariance is the formula
20	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
21	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
22	Uncorrelated noise	none: the covariance is the formula
23	A rough function with cubicly varying marginal standard deviation	none: the covariance is the formula
24	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
25	A rough function with cubicly varying marginal standard deviation	none: the covariance is the formula
26	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
27	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
28	A rough function with quarticly varying marginal standard deviation	none: the covariance is the formula
29	A rough function with cubicly varying marginal standard deviation	none: the covariance is the formula
30	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
31	A rough function with cubicly varying marginal standard deviation	none: the covariance is the formula
32	A rough function with marginal standard deviation increasing linearly away from Jun 1954	none: the covariance is the formula
33	A rough function with quadraticly varying marginal standard deviation	none: the covariance is the formula
34	A rough function with cubicly varying marginal standard deviation	none: the covariance is the formula
35	A rough function with marginal standard	none: the covariance is the formula



Data and full posterior

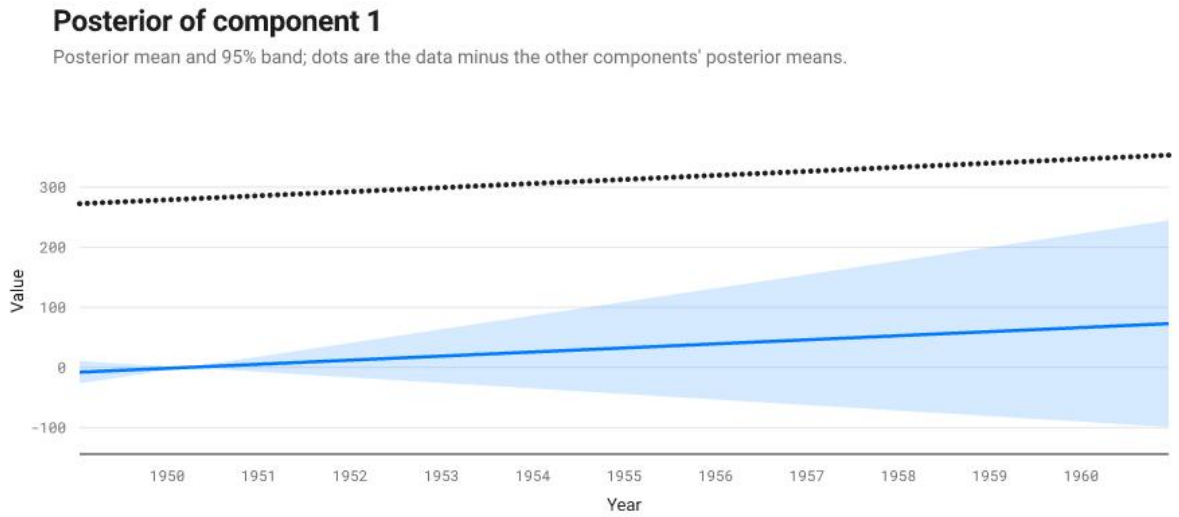
Detailed discussion of the components

Component 1: A linearly increasing function

This component is linearly increasing.

This component explains 32.4% of the data's variance on its own; with the components above it, 32.4%. Adding it changes the cross-validated MAE by -12.0%.

Extrapolation. This component is assumed to continue to increase linearly.



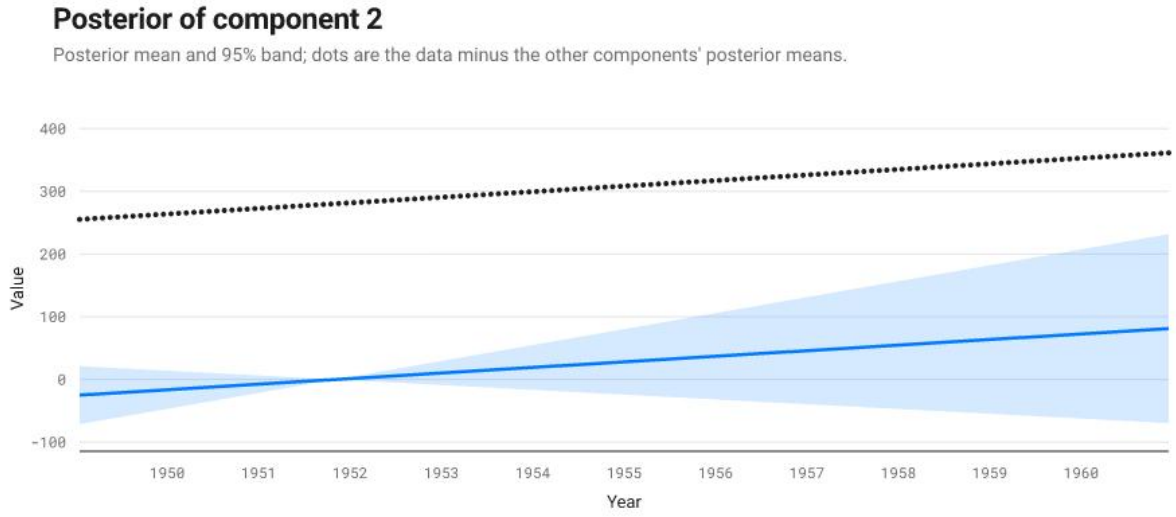
Component 1

Component 2: A linearly increasing function

This component is linearly increasing.

This component explains 41.0% of the data's variance on its own; with the components above it, 63.3%. Adding it changes the cross-validated MAE by -11.1%.

Extrapolation. This component is assumed to continue to increase linearly.



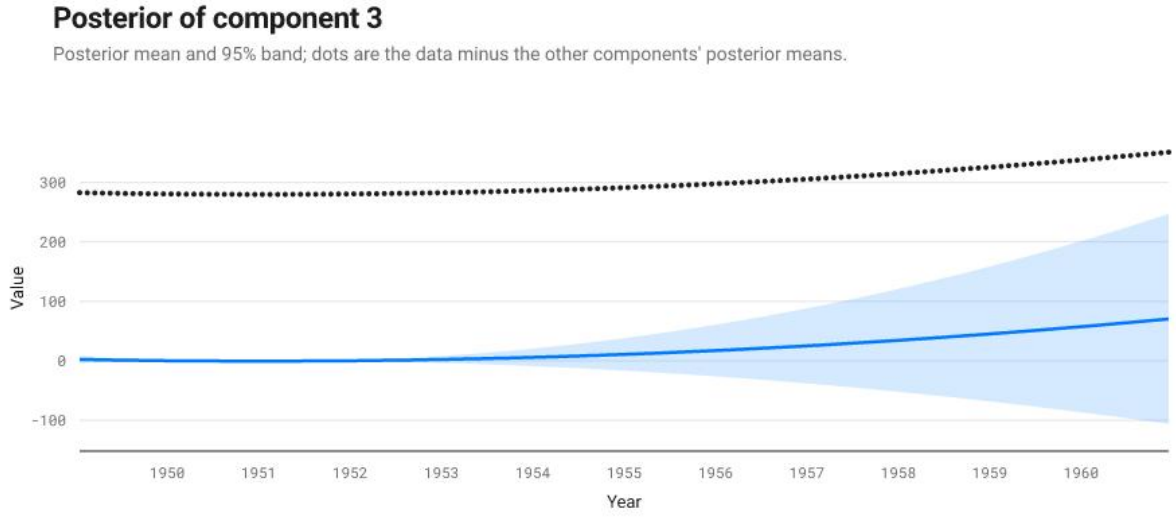
Component 2

Component 3: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 28.6% of the data's variance on its own; with the components above it, 76.9%. Adding it changes the cross-validated MAE by -8.7%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



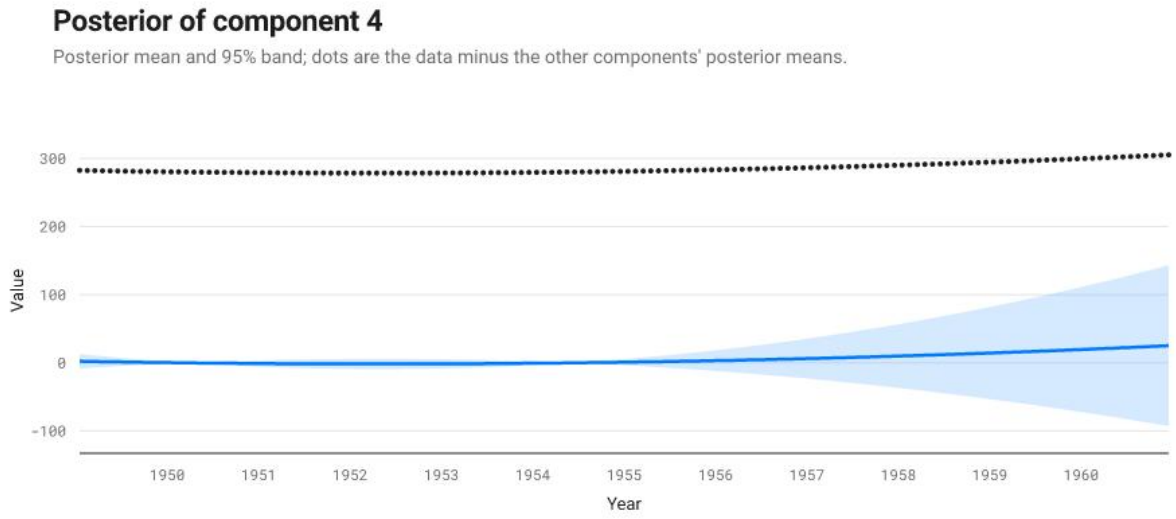
Component 3

Component 4: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 10.4% of the data's variance on its own; with the components above it, 80.0%. Adding it changes the cross-validated MAE by -2.5%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



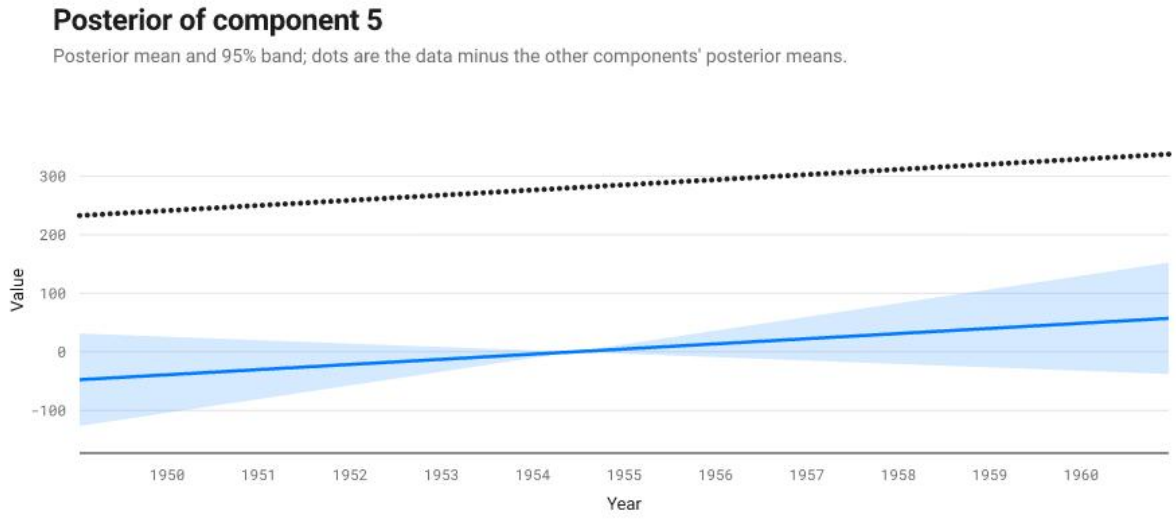
Component 4

Component 5: A linearly increasing function

This component is linearly increasing.

This component explains 40.6% of the data's variance on its own; with the components above it, 86.2%. Adding it changes the cross-validated MAE by -2.1%.

Extrapolation. This component is assumed to continue to increase linearly.



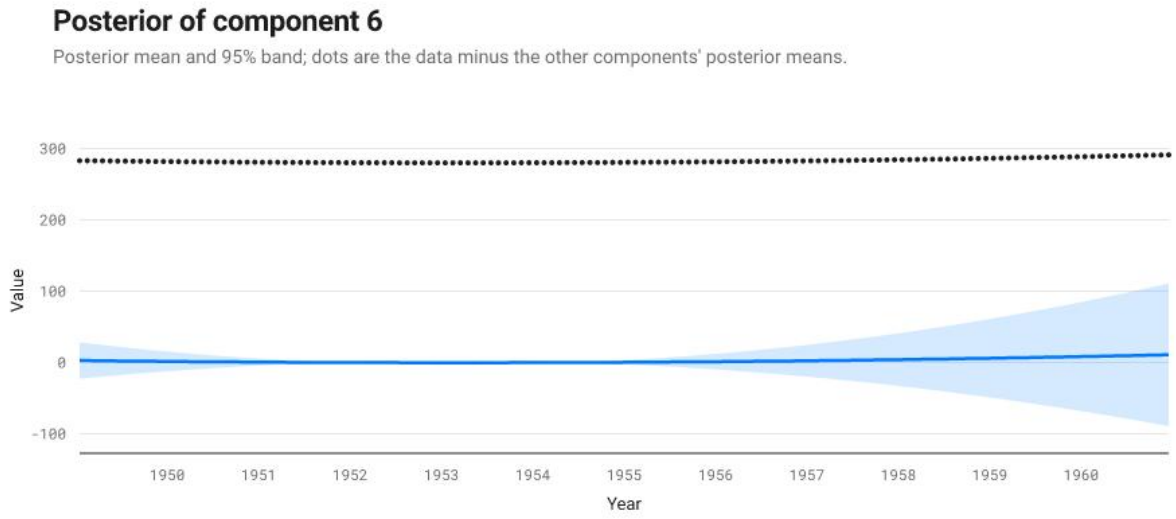
Component 5

Component 6: A quadratic polynomial

This component is a quadratic polynomial.

This component explains 3.9% of the data's variance on its own; with the components above it, 86.1%. Adding it changes the cross-validated MAE by -1.2%.

Extrapolation. This component is assumed to continue as a quadratic polynomial.



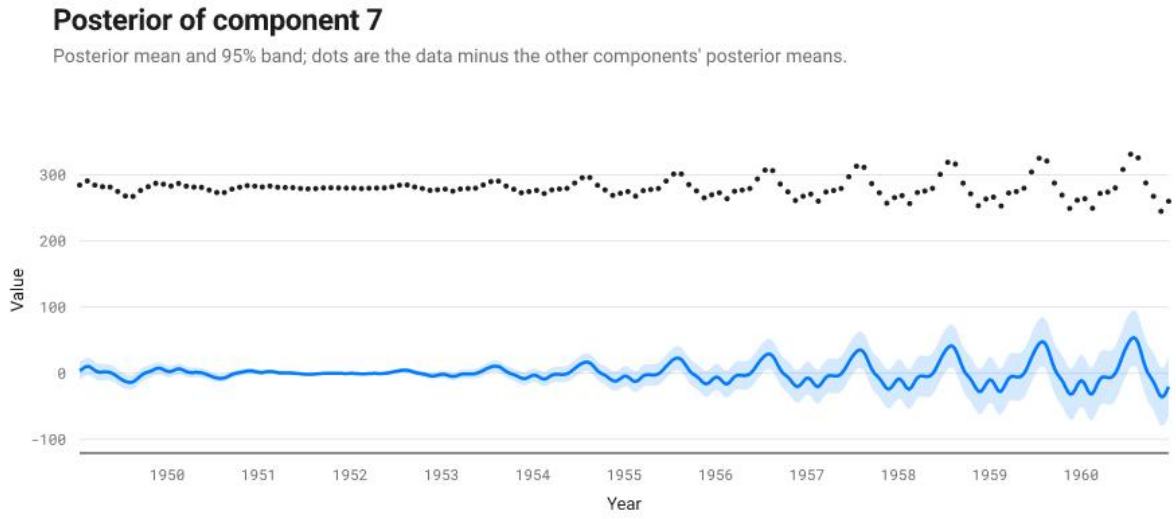
Component 6

Component 7: A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Nov 1951

This component is periodic with a period of 1.0 years but with varying amplitude. The amplitude of the function increases linearly away from Nov 1951. The shape of this function within each period has a typical lengthscale of 4.0 weeks.

This component explains 6.8% of the data's variance on its own; with the components above it, 92.8%. Adding it changes the cross-validated MAE by -0.2%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years but with varying amplitude. The amplitude of the function is assumed to continue to increase linearly.



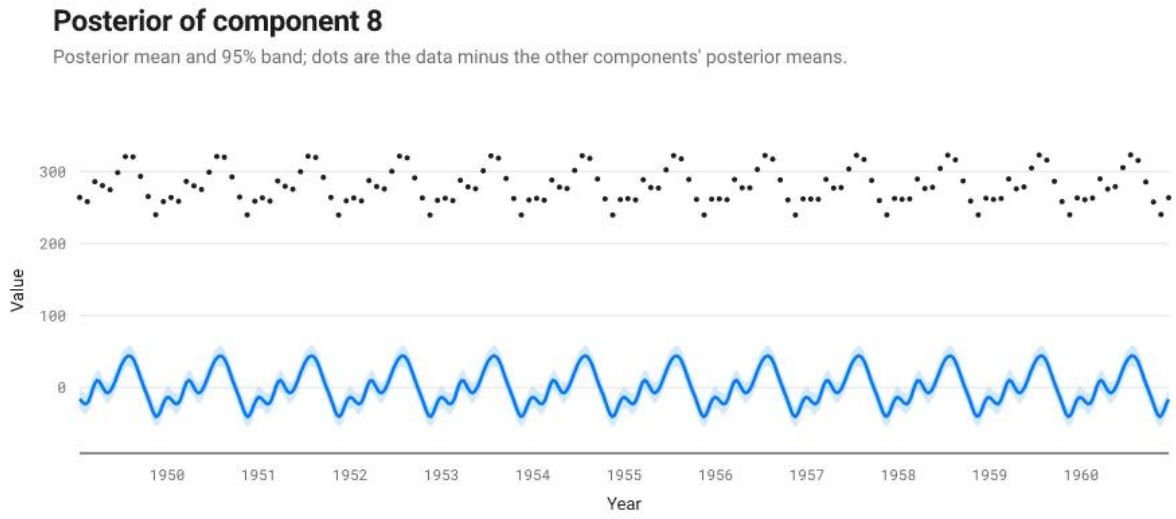
Component 7

Component 8: A periodic function with a period of 1.0 years

This component is periodic with a period of 1.0 years. The shape of this function within each period has a typical lengthscale of 4.0 weeks.

This component explains 8.3% of the data's variance on its own; with the components above it, 98.2%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years.



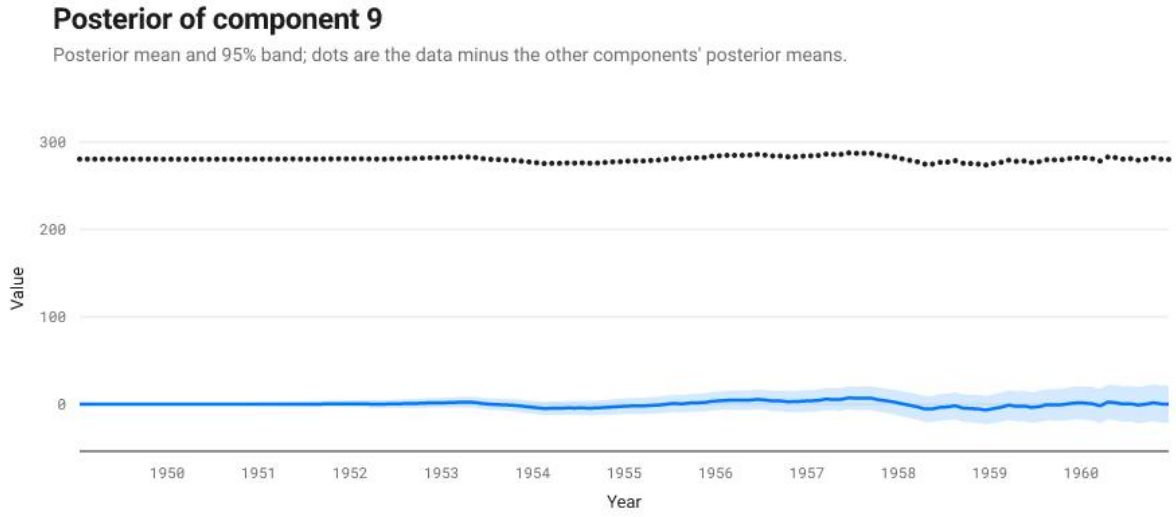
Component 8

Component 9: A rough function with marginal standard deviation increasing linearly away from Feb 1950

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function increases linearly away from Feb 1950. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains 0.3% of the data's variance on its own; with the components above it, 98.6%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to increase linearly.



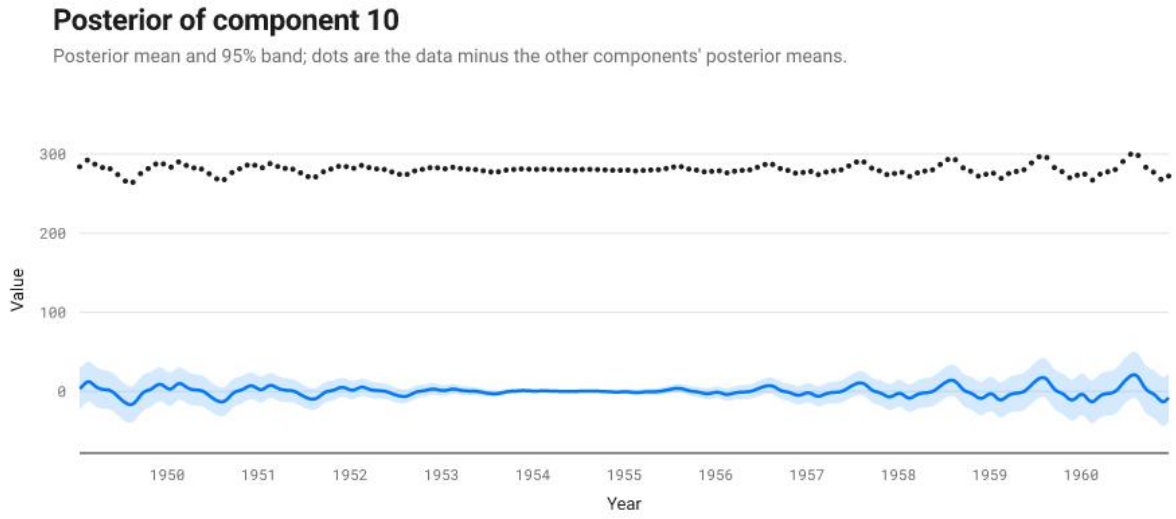
Component 9

Component 10: A periodic function with a period of 1.0 years but with amplitude increasing linearly away from Jun 1954

This component is periodic with a period of 1.0 years but with varying amplitude. The amplitude of the function increases linearly away from Jun 1954. The shape of this function within each period has a typical lengthscale of 4.0 weeks.

This component explains 1.9% of the data's variance on its own; with the components above it, 99.2%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years but with varying amplitude. The amplitude of the function is assumed to continue to increase linearly.



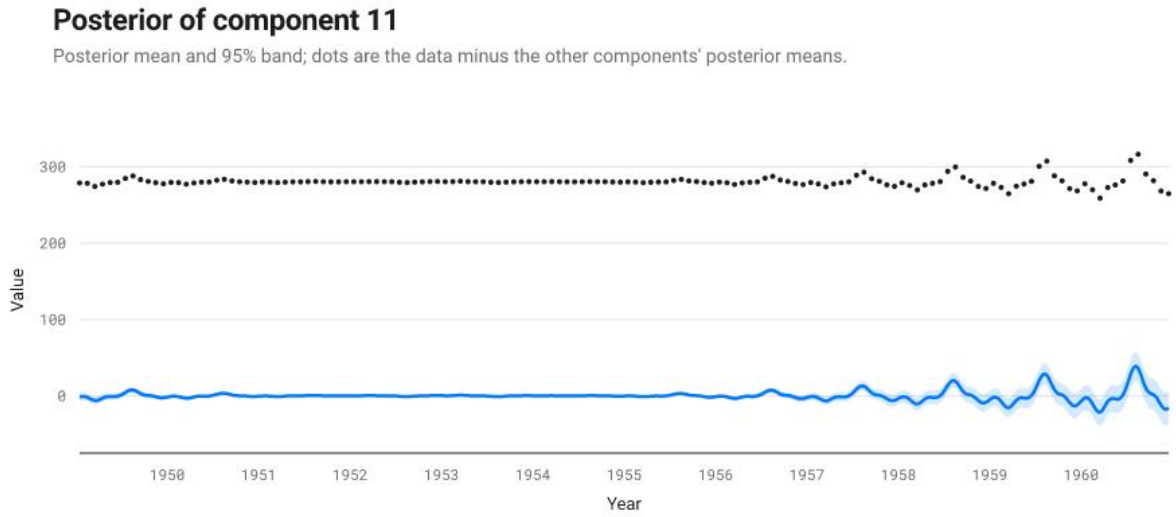
Component 10

Component 11: A periodic function with a period of 1.0 years but with quadraticly varying amplitude

This component is periodic with a period of 1.0 years but with varying amplitude. The amplitude of the function varies quadraticly. The shape of this function within each period has a typical lengthscale of 4.0 weeks.

This component explains 3.1% of the data's variance on its own; with the components above it, 99.5%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years but with varying amplitude. The amplitude of the function is assumed to continue to vary quadraticly.



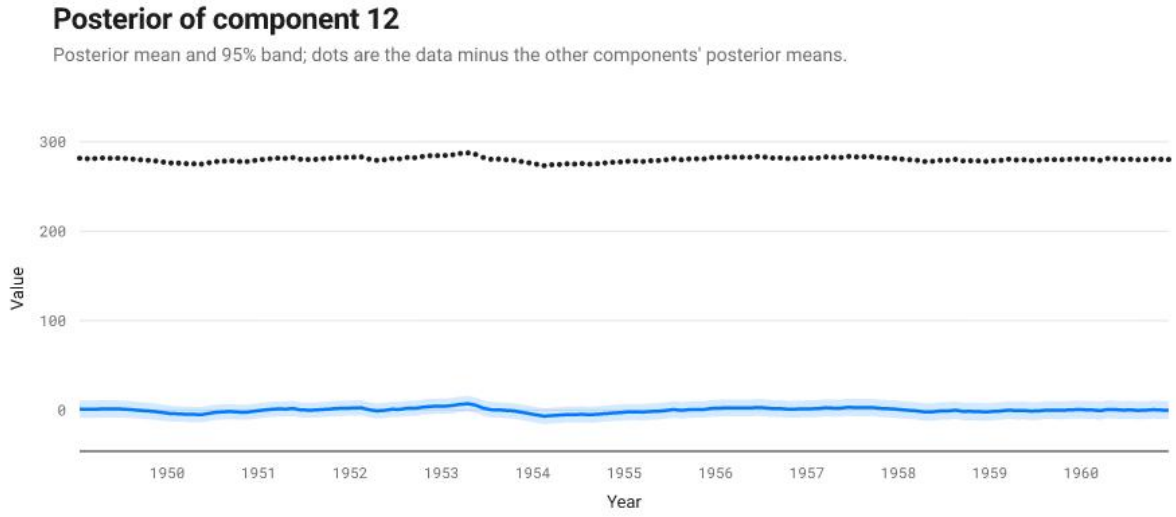
Component 11

Component 12: A rough function

This component is a rough function with a typical lengthscale of 4.2 months. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.52).

This component explains 0.5% of the data's variance on its own; with the components above it, 99.7%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.2 months.



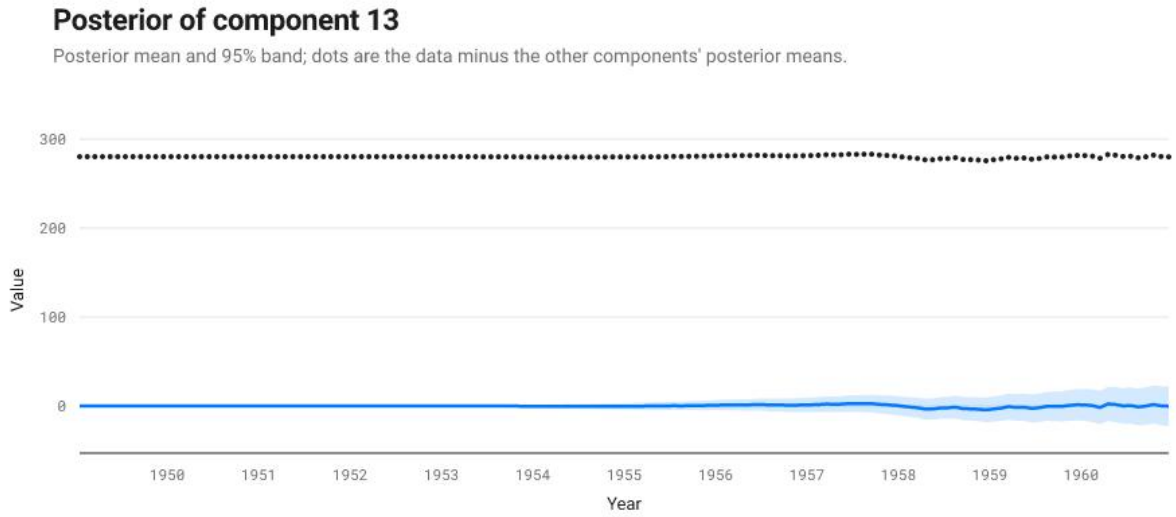
Component 12

Component 13: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies quadraticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.7%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



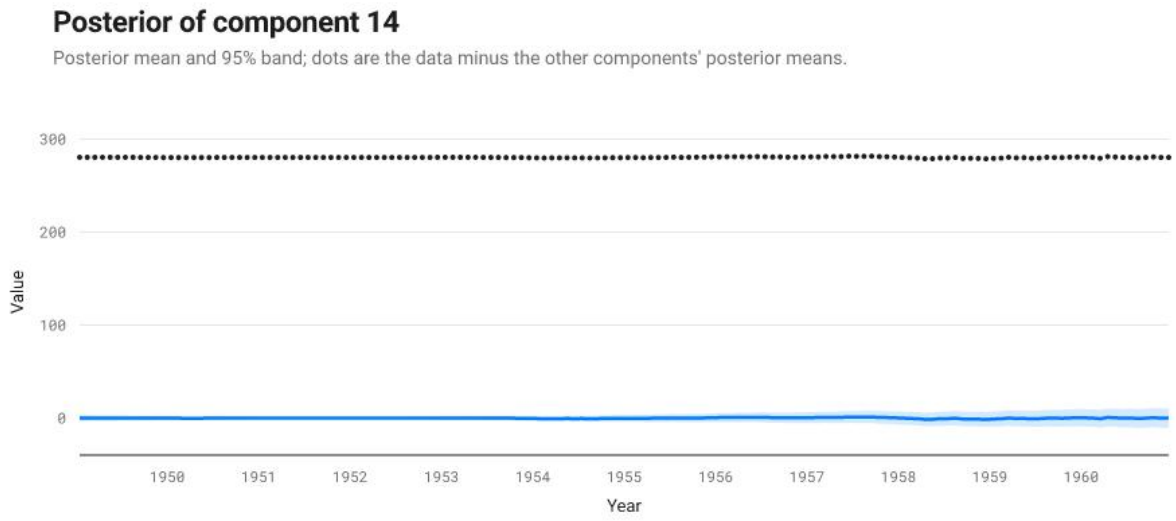
Component 13

Component 14: A rough function with marginal standard deviation increasing linearly away from Nov 1951

This component is a rough function with a typical lengthscale of 4.2 months. The marginal standard deviation of the function increases linearly away from Nov 1951. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.52).

This component explains 0.1% of the data's variance on its own; with the components above it, 99.8%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.2 months. The marginal standard deviation of the function is assumed to continue to increase linearly.



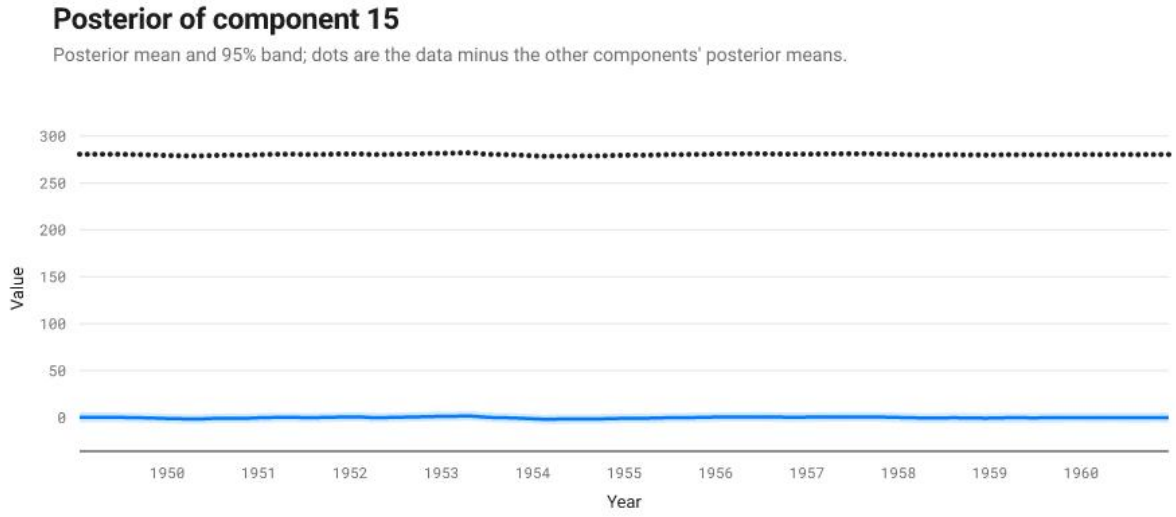
Component 14

Component 15: A rough function

This component is a rough function with a typical lengthscale of 4.3 months. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains 0.2% of the data's variance on its own; with the components above it, 99.8%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months.



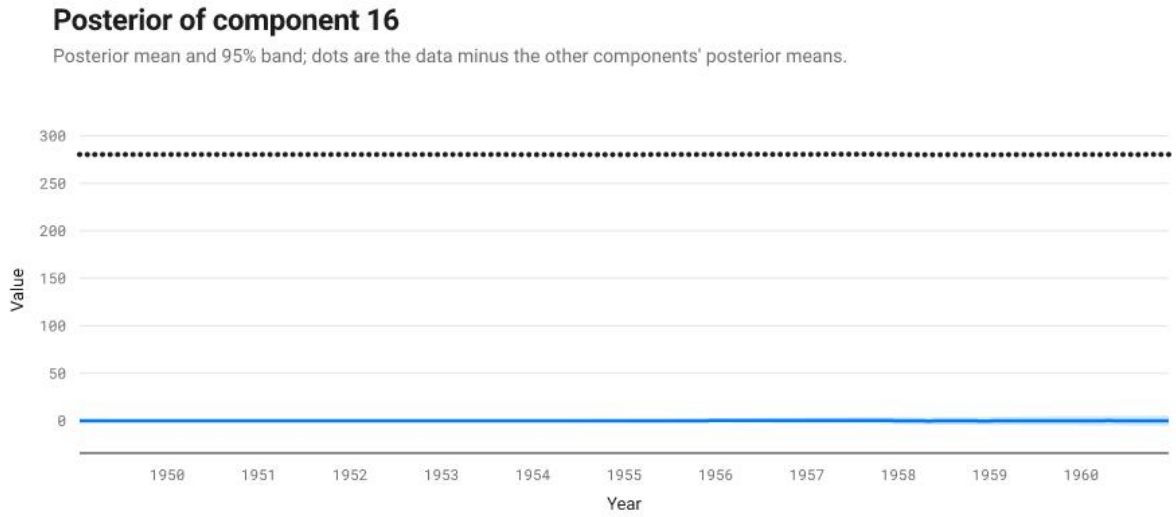
Component 15

Component 16: A rough function with marginal standard deviation increasing linearly away from Nov 1951

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function increases linearly away from Nov 1951. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains 0.0% of the data's variance on its own; with the components above it, 99.8%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to increase linearly.



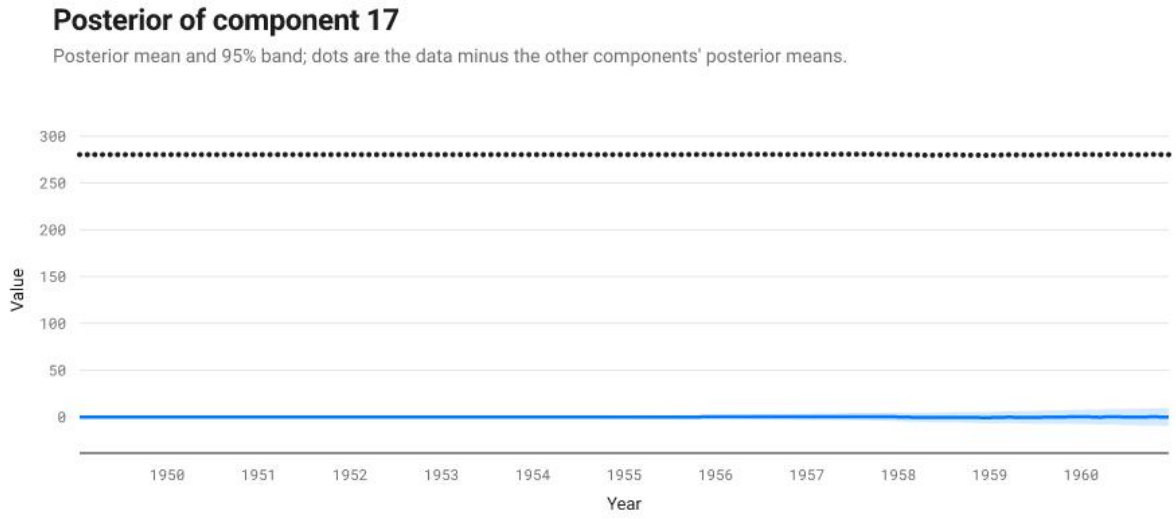
Component 16

Component 17: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies quadratically. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.8%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary quadratically.



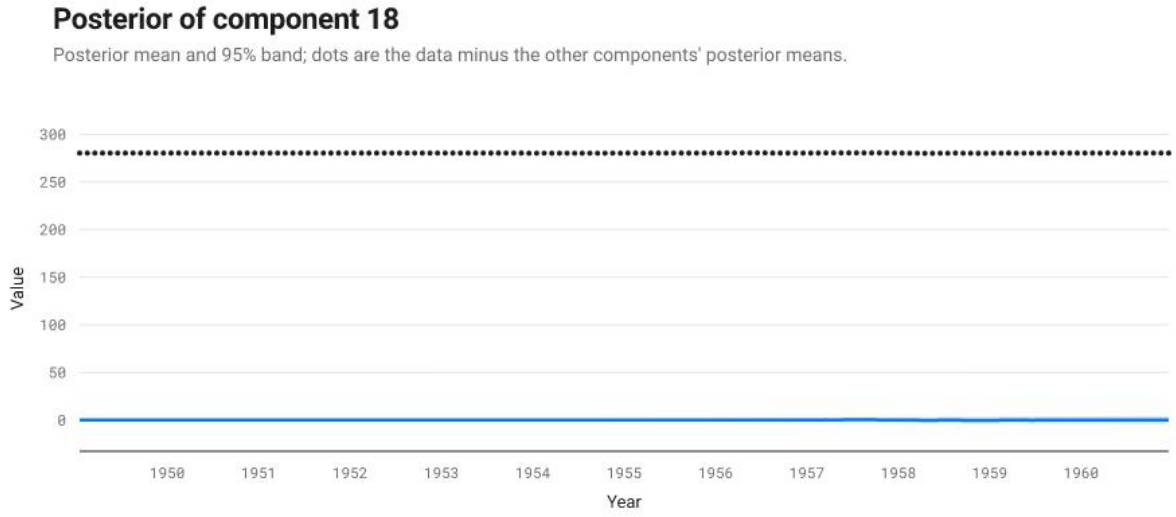
Component 17

Component 18: A rough function with marginal standard deviation increasing linearly away from Dec 1951

This component is a rough function with a typical lengthscale of 4.2 months. The marginal standard deviation of the function increases linearly away from Dec 1951. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.52).

This component explains 0.0% of the data's variance on its own; with the components above it, 99.8%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.2 months. The marginal standard deviation of the function is assumed to continue to increase linearly.



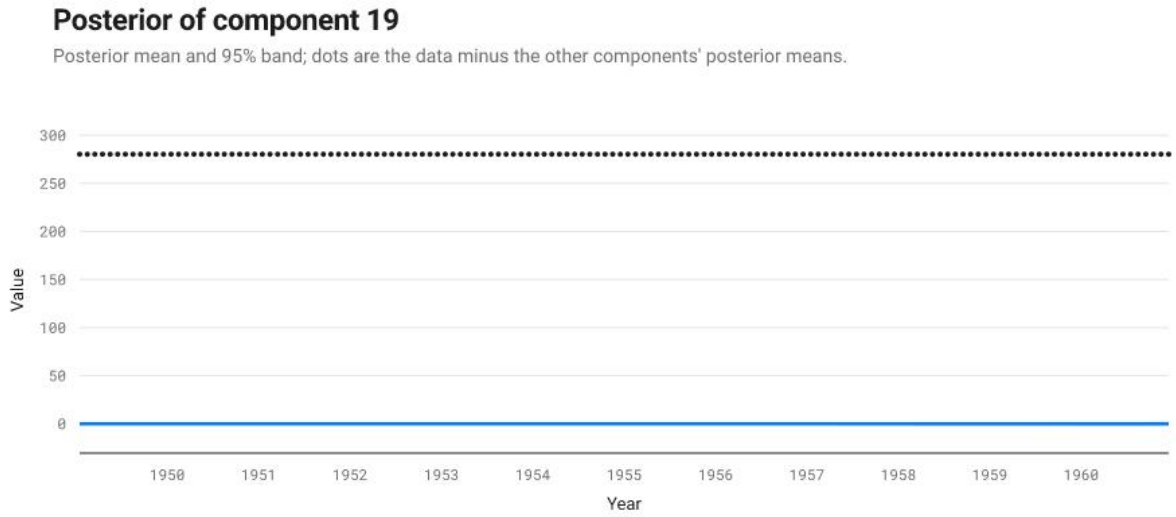
Component 18

Component 19: A rough function with marginal standard deviation increasing linearly away from Dec 1951

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function increases linearly away from Dec 1951. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains 0.0% of the data's variance on its own; with the components above it, 99.8%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to increase linearly.



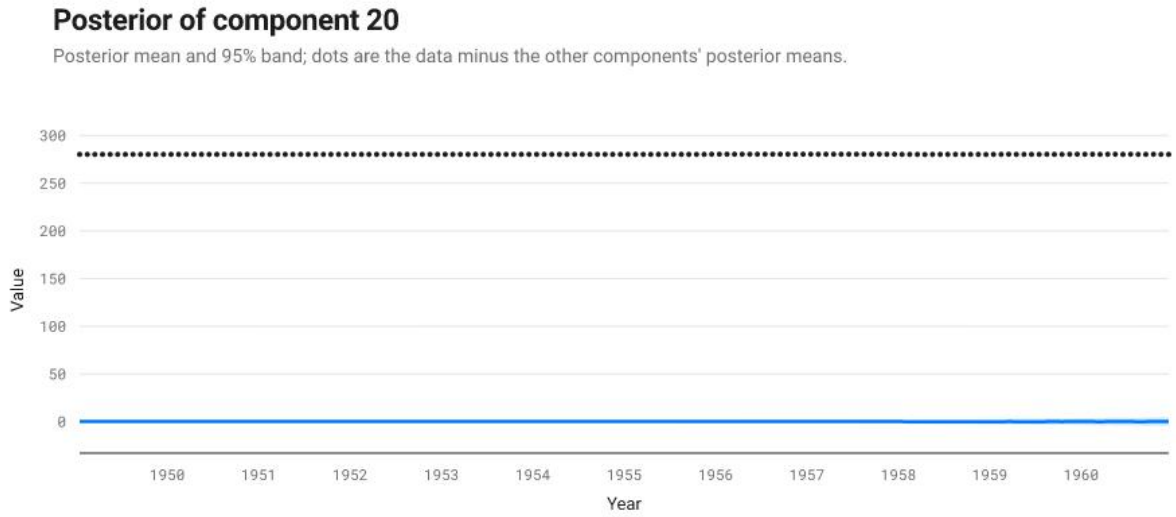
Component 19

Component 20: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.2 months. The marginal standard deviation of the function varies quadraticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.52).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.8%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.2 months. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



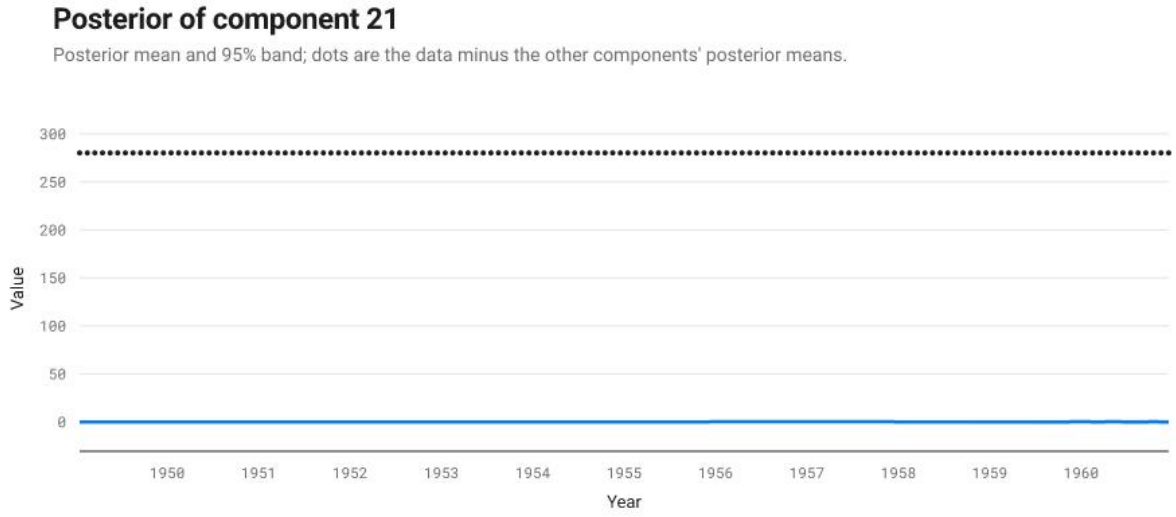
Component 20

Component 21: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies quadraticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.8%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



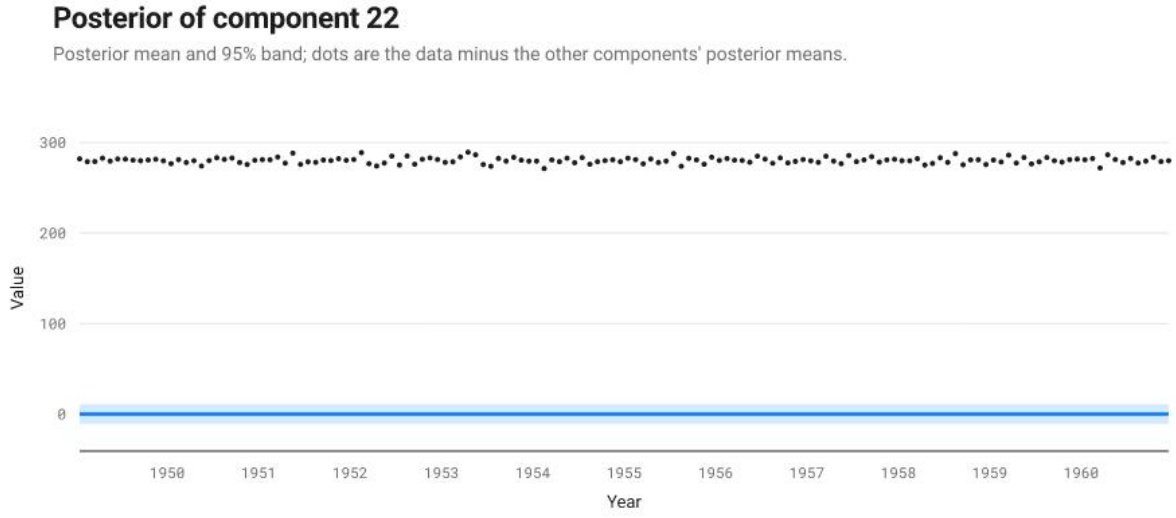
Component 21

Component 22: Uncorrelated noise

This component models uncorrelated noise.

This component explains 0.3% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component assumes the uncorrelated noise will continue indefinitely.



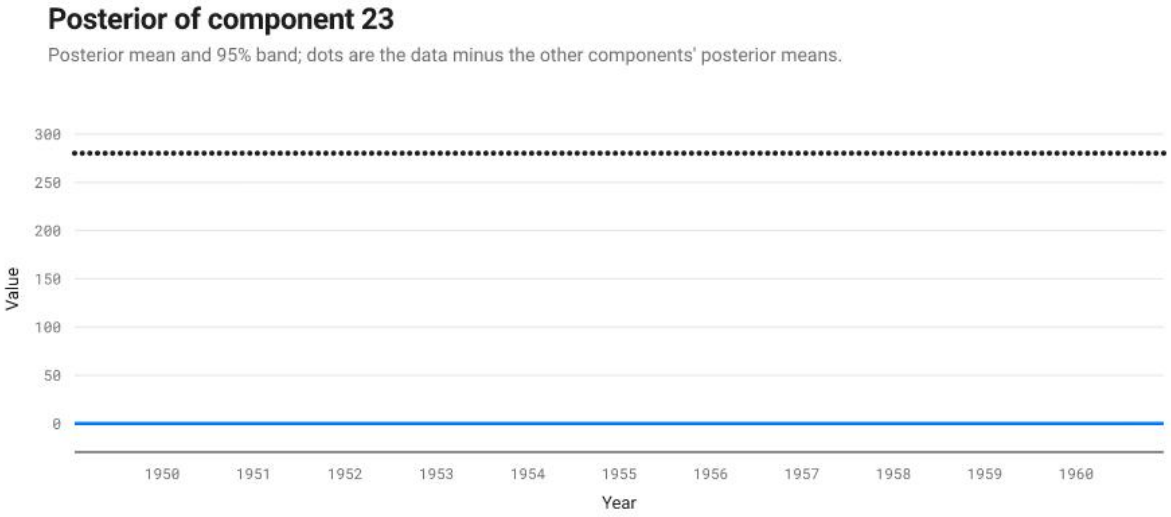
Component 22

Component 23: A rough function with cubicly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies cubicly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary cubicly.



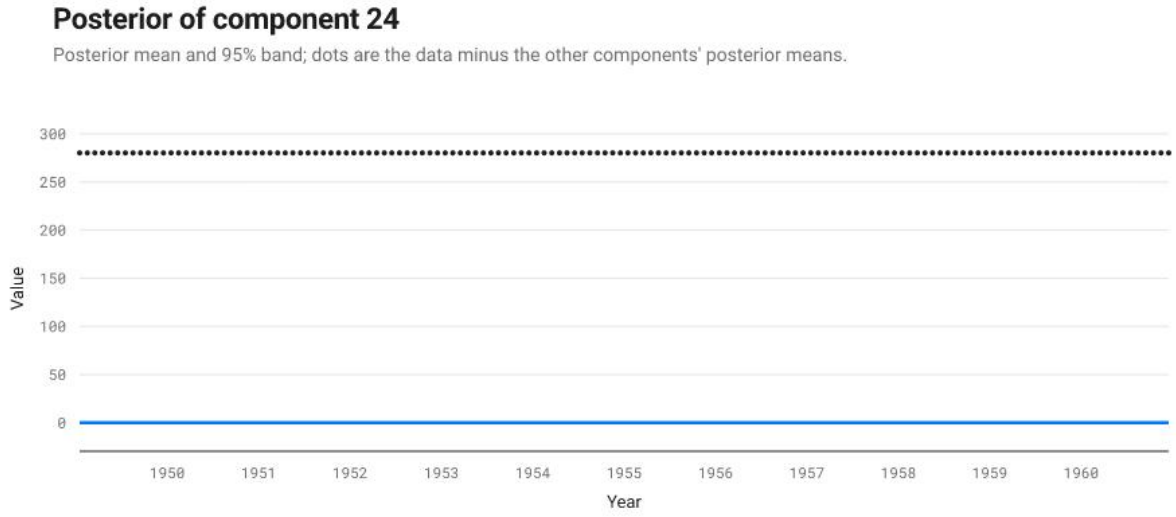
Component 23

Component 24: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies quadratically. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary quadratically.



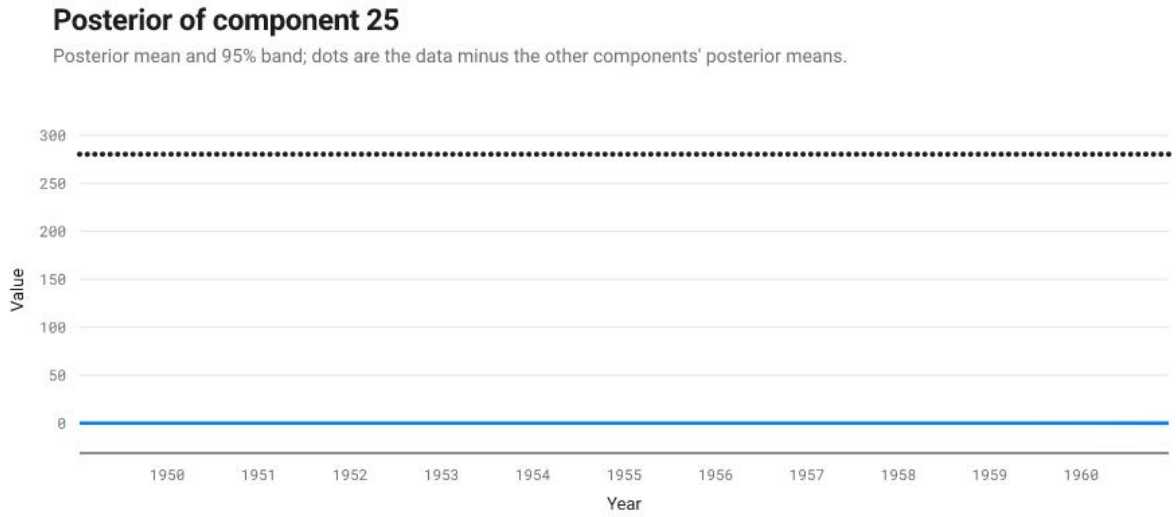
Component 24

Component 25: A rough function with cubicly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.2 months. The marginal standard deviation of the function varies cubicly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.52).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.2 months. The marginal standard deviation of the function is assumed to continue to vary cubicly.



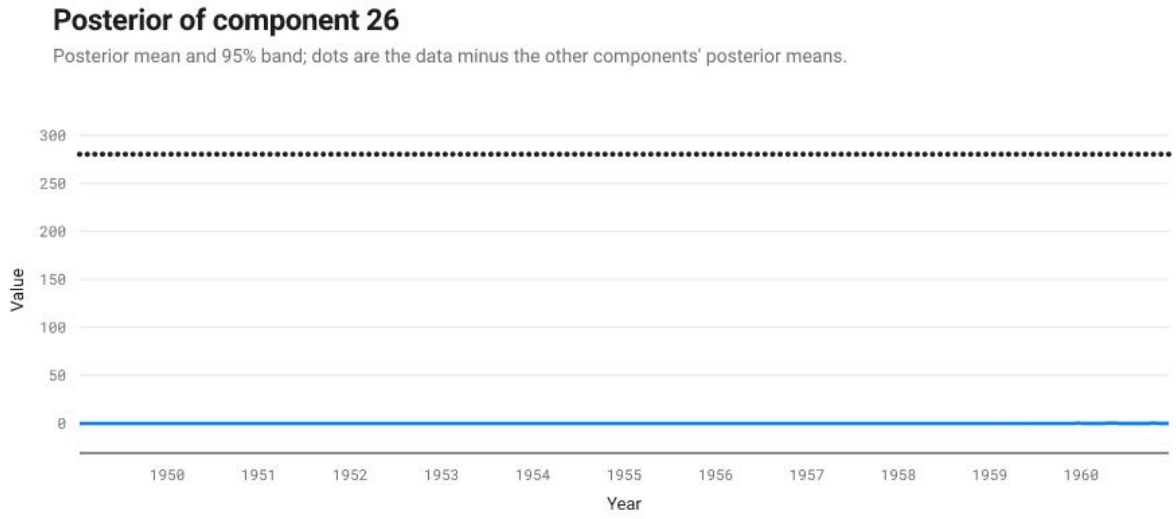
Component 25

Component 26: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.2 months. The marginal standard deviation of the function varies quadraticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.52).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.2 months. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



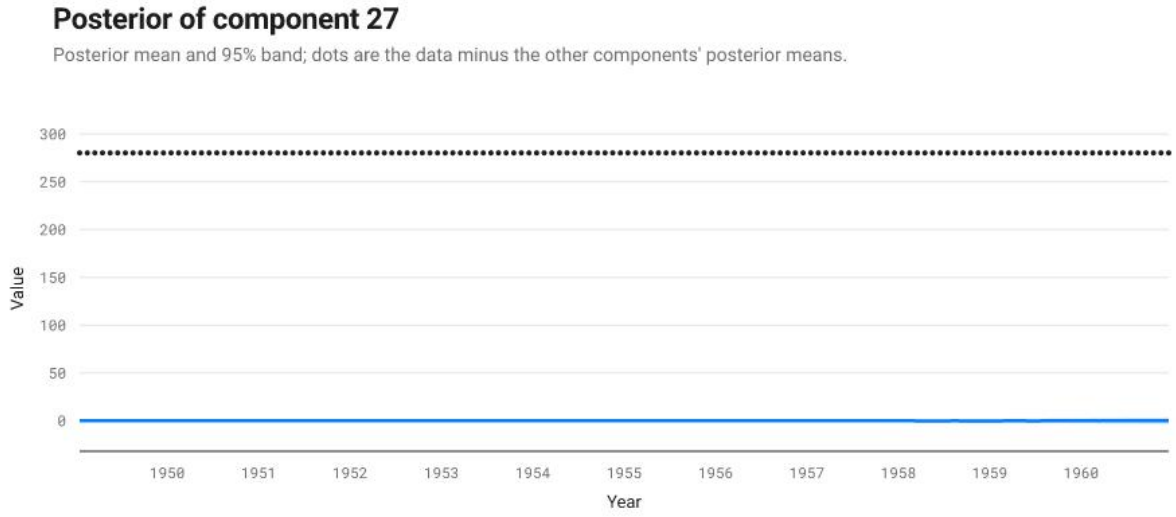
Component 26

Component 27: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies quadraticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



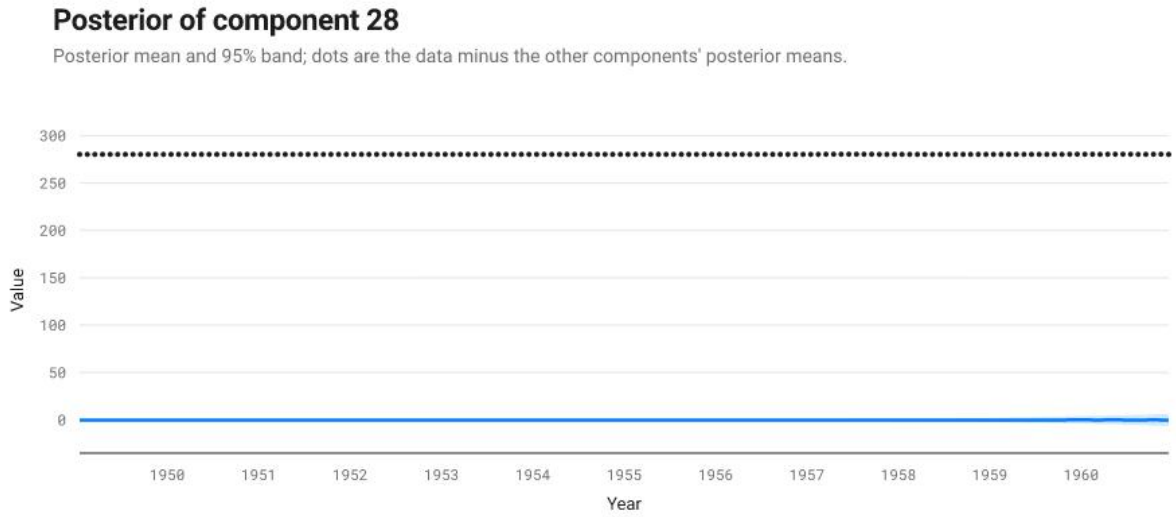
Component 27

Component 28: A rough function with quarticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies quarticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary quarticly.



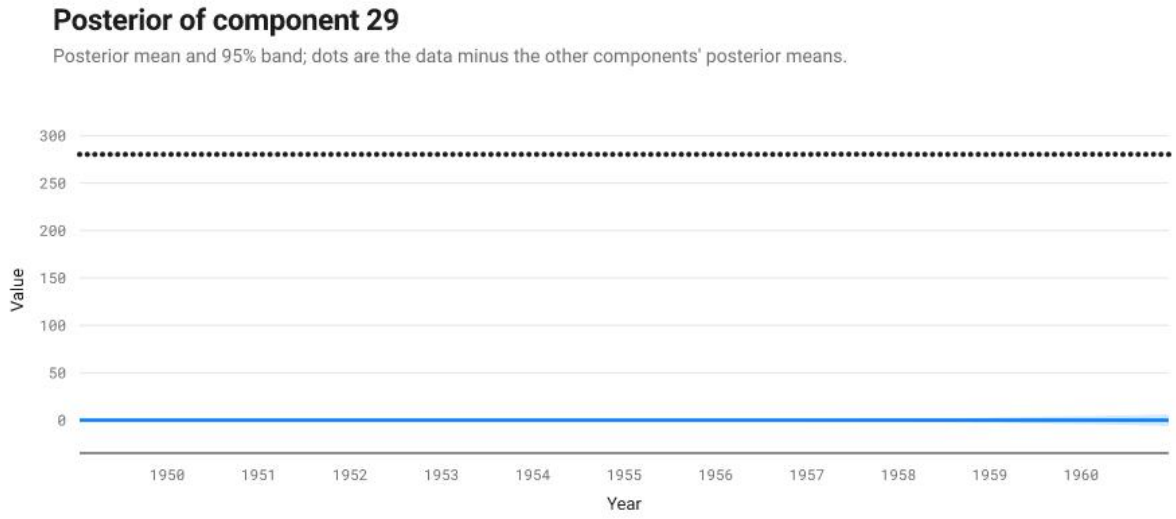
Component 28

Component 29: A rough function with cubicly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies cubicly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary cubicly.



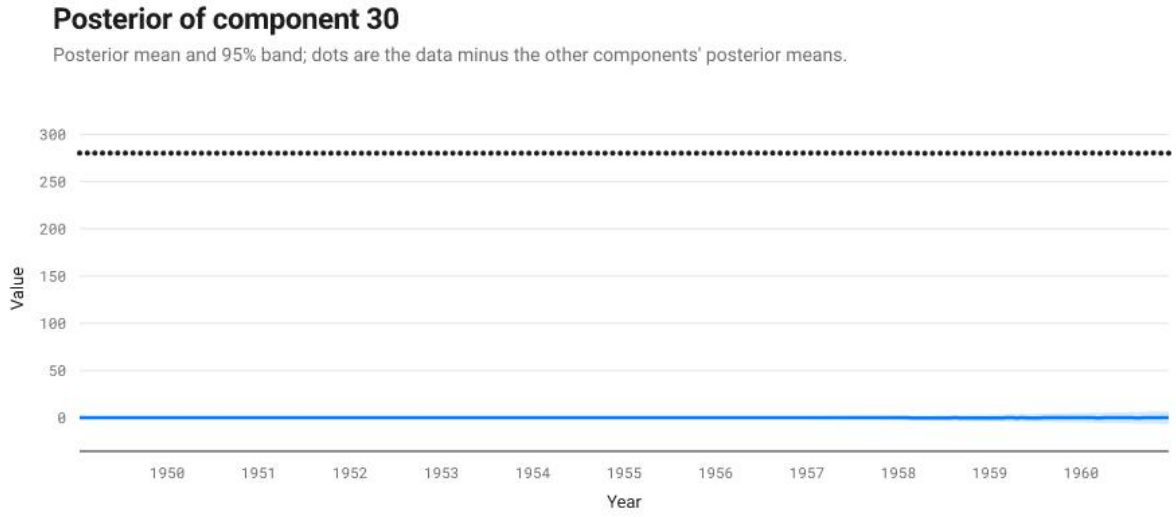
Component 29

Component 30: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.2 months. The marginal standard deviation of the function varies quadraticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.52).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.2 months. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



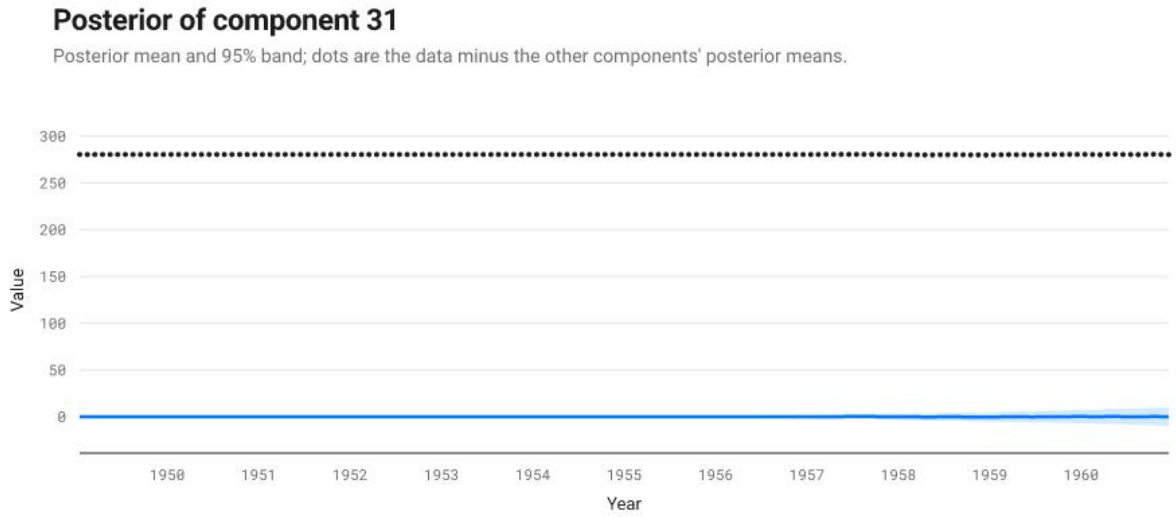
Component 30

Component 31: A rough function with cubicly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies cubicly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary cubicly.



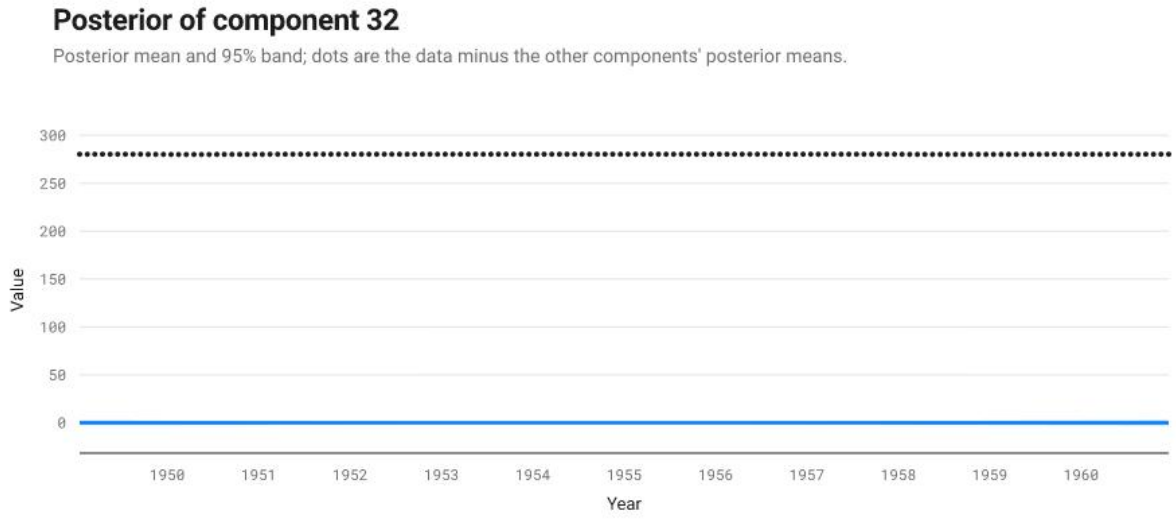
Component 31

Component 32: A rough function with marginal standard deviation increasing linearly away from Jun 1954

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function increases linearly away from Jun 1954. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains 0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to increase linearly.



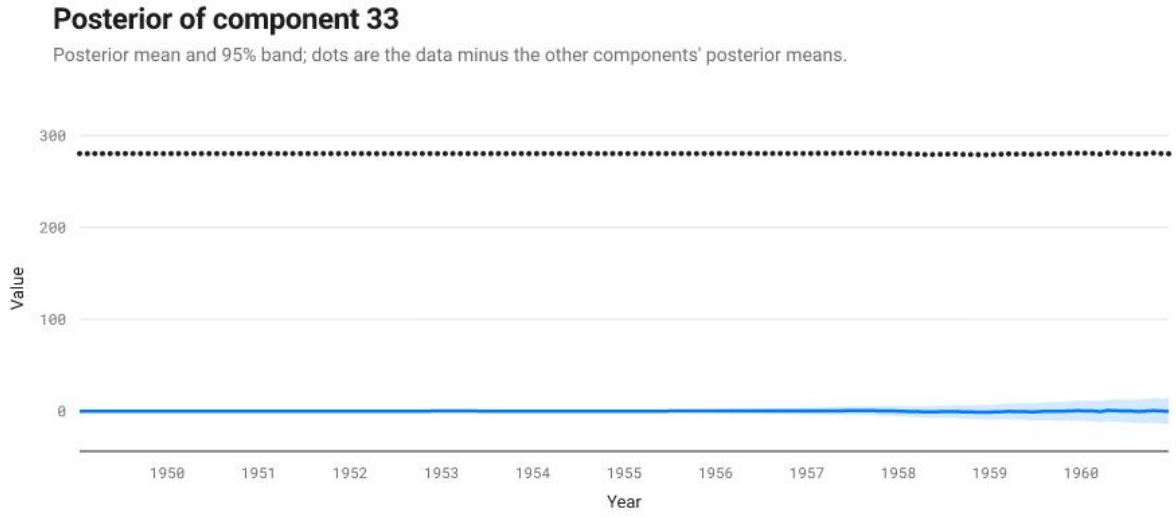
Component 32

Component 33: A rough function with quadraticly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies quadraticly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary quadraticly.



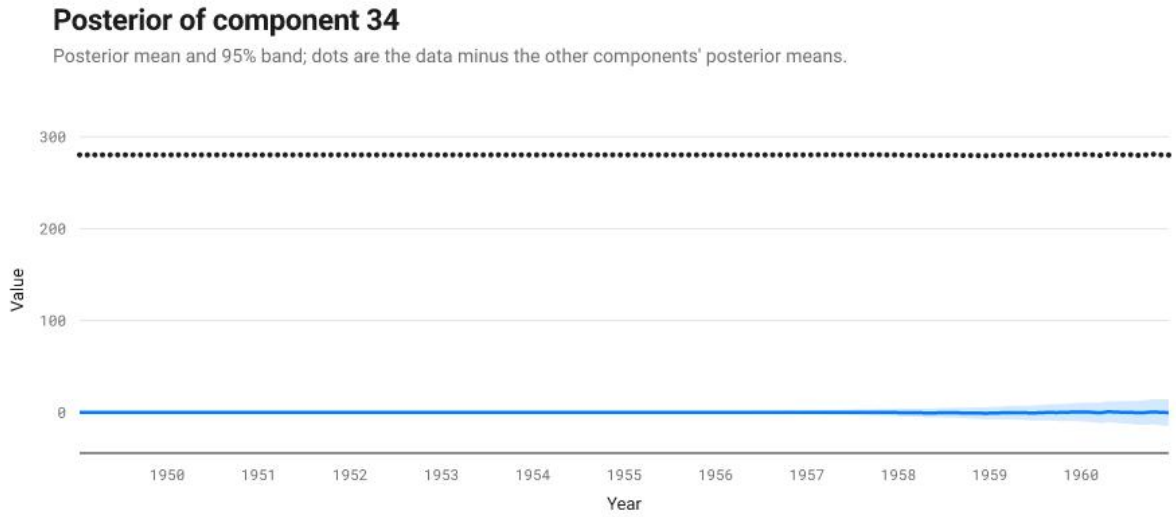
Component 33

Component 34: A rough function with cubicly varying marginal standard deviation

This component is a rough function with a typical lengthscale of 4.3 months. The marginal standard deviation of the function varies cubicly. Its sample paths are continuous but not differentiable (gamma-exponential exponent 1.03).

This component explains -0.0% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.3 months. The marginal standard deviation of the function is assumed to continue to vary cubicly.



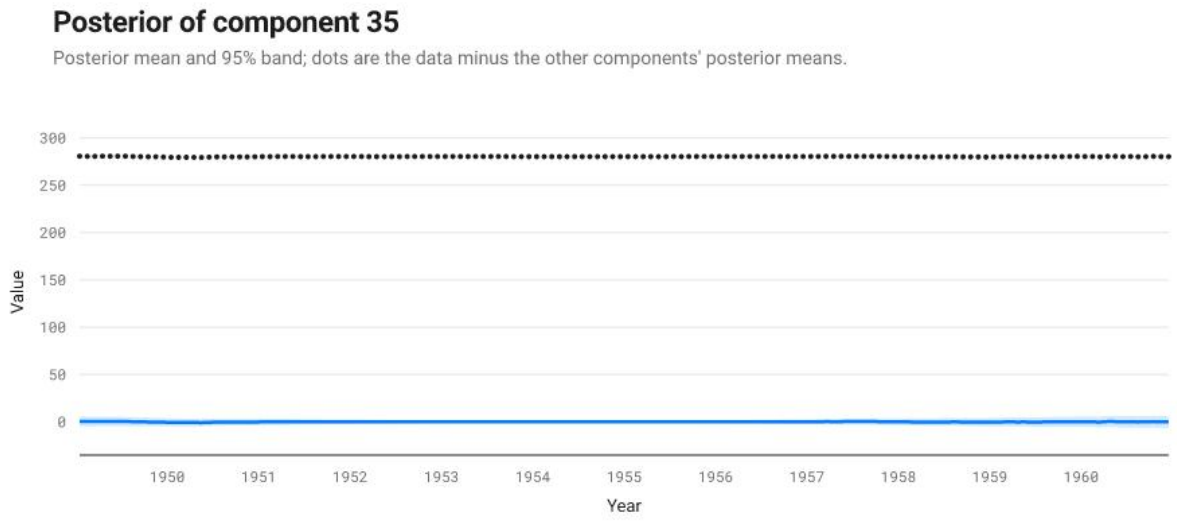
Component 34

Component 35: A rough function with marginal standard deviation increasing linearly away from Jun 1954

This component is a rough function with a typical lengthscale of 4.2 months. The marginal standard deviation of the function increases linearly away from Jun 1954. Its sample paths are continuous but not differentiable (gamma-exponential exponent 0.52).

This component explains 0.1% of the data's variance on its own; with the components above it, 99.9%. Adding it changes the cross-validated MAE by 0.0%.

Extrapolation. This component is assumed to continue roughly but is also assumed to be stationary so its distribution will return to the prior. The prior distribution places mass on rough functions with a marginal mean of zero and a typical lengthscale of 4.2 months. The marginal standard deviation of the function is assumed to continue to increase linearly.



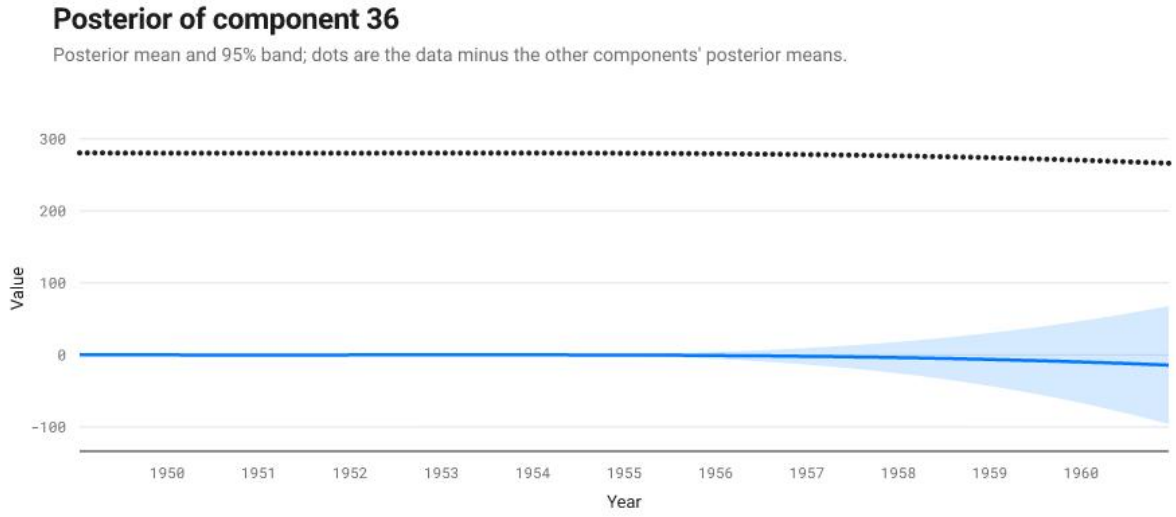
Component 35

Component 36: A cubic polynomial

This component is a cubic polynomial.

This component explains -5.3% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by 1.5%.

Extrapolation. This component is assumed to continue as a cubic polynomial.



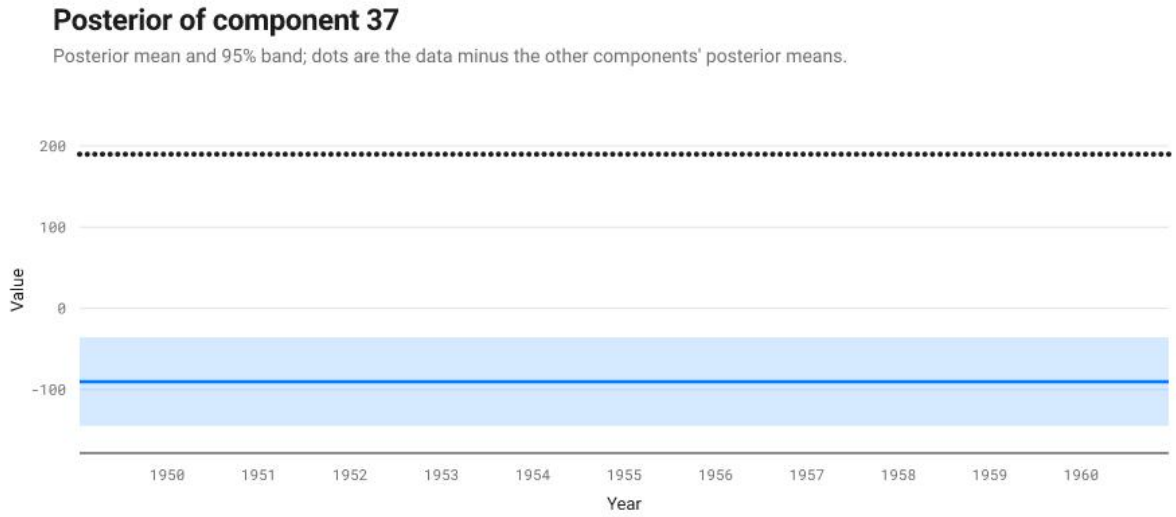
Component 36

Component 37: A constant

This component is constant.

This component explains -0.0% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by 47.4%.

Extrapolation. This component is assumed to stay constant.



Component 37

Model checking

Each statistic compares draws of a component's posterior with draws of its prior. A small p-value (bold below 0.05) points at a property of the data that the component's prior does not reproduce.

#	QQ max	QQ min	ACF min	ACF min lag	Pgram max	Pgram peak	MMD
1	0.790	0.676	0.982	0.988	0.950	0.974	**0.000**
2	0.842	0.576	0.976	0.966	0.806	0.998	**0.001**
3	0.742	0.800	0.980	0.994	0.998	0.998	**0.004**
4	0.818	0.850	0.988	0.970	0.986	0.980	0.259
5	0.960	0.718	0.936	0.984	0.792	0.904	0.130
6	0.918	0.976	0.922	0.948	0.844	0.980	**0.001**
7	0.904	0.770	0.844	0.916	0.754	0.784	**0.000**
8	0.528	0.676	0.922	0.992	0.716	0.482	**0.000**
9	0.938	0.962	0.820	0.958	0.926	0.950	0.605
10	0.966	0.974	0.872	0.958	0.892	0.848	0.111
11	0.630	0.448	0.916	0.942	0.896	0.598	0.756
12	0.970	0.984	0.902	0.932	0.882	0.928	0.682
13	0.990	0.978	0.964	0.974	1.000	0.960	0.389
14	0.988	0.998	0.950	0.988	0.996	0.972	0.061
15	0.988	0.938	0.970	0.936	0.978	0.930	0.749
16	0.972	0.984	0.994	0.958	0.966	0.958	0.503
17	0.988	0.998	0.942	0.992	0.992	0.976	0.287
18	0.996	0.984	0.970	0.990	1.000	0.966	0.561
19	0.986	0.990	0.976	0.982	0.978	0.960	0.704
20	0.950	0.930	0.972	0.982	0.964	0.926	0.565
21	0.956	0.946	0.962	0.976	0.980	0.988	0.827
22	0.994	0.914	0.958	0.954	0.962	0.986	**0.003**
23	0.968	0.978	0.994	0.998	0.984	0.994	0.500
24	0.972	0.976	0.994	0.970	0.982	0.996	0.375
25	0.958	0.970	0.948	0.944	0.990	0.952	0.716
26	0.976	0.934	0.988	0.960	0.994	0.988	0.443
27	0.998	0.982	0.988	0.982	0.992	0.986	0.586
28	0.982	0.994	0.974	1.000	0.952	0.966	0.916
29	0.988	0.968	0.974	0.998	0.982	0.980	0.789
30	0.988	0.968	0.978	1.000	0.982	0.970	0.425
31	0.984	0.966	0.992	0.964	0.992	0.998	0.914
32	0.968	0.942	0.976	0.984	0.986	0.994	0.860
33	0.976	0.948	0.944	0.988	0.998	0.934	0.097
34	0.942	0.990	0.994	0.972	0.958	0.998	0.682
35	0.972	0.966	0.974	0.950	0.974	0.944	0.261
36	0.950	0.680	0.986	0.994	0.656	0.986	**0.000**
37	0.388	0.388	0.810	0.646	0.682	0.972	**0.000**

QQ: quantile-quantile gaps. ACF: autocorrelation function. Pgram: periodogram. MMD: maximum mean discrepancy two-sample test.